

Rates Fundamentals

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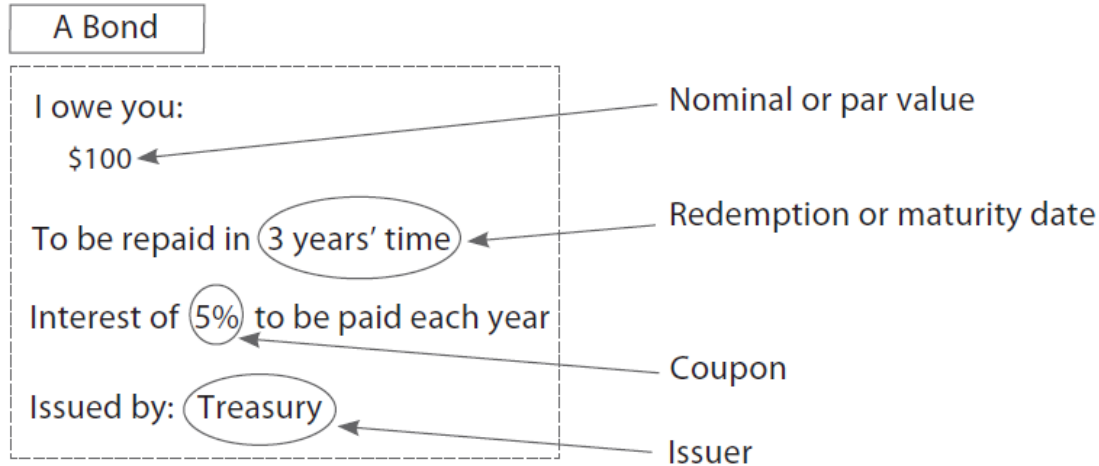
- Fixed income basics
- Post-08 changes in monetary policy
- Different “bets” in rates markets

Fixed income basics

Fixed Income

Basic Features

1. Issuer
2. Nominal value (or par value)
3. Coupon
4. Redemption/maturity date



Pricing

- Like any asset, fixed income is priced by discounting cashflows with the risk-free interest rate
- When interest rate rises, cashflows are discounted more stringently
 - New bonds pay higher interest rate, so old (i.e. existing) bonds become less popular and so cheaper

$$B_0 = \frac{C}{(1+R)^1} + \frac{C}{(1+R)^2} + \dots + \frac{C+P}{(1+R)^n}$$

B_0 = price of bond

C = coupon payment

R = discount rate (yield to maturity)

P = bond's par value

n = number of coupon payments

Duration & Convexity

- **Duration**: a number that tells you – how sensitive is a bond to changes in interest rates?
- **Convexity**: positive vs. negative – positive convexity is asymmetric upside for the investor – when the bond's price rises when its yield falls more than when its price falls when its yield rises

Rates vs. Credit

Rates

What's the level of interest rate?

Bet on Interest Rate Risk

Macro

Credit

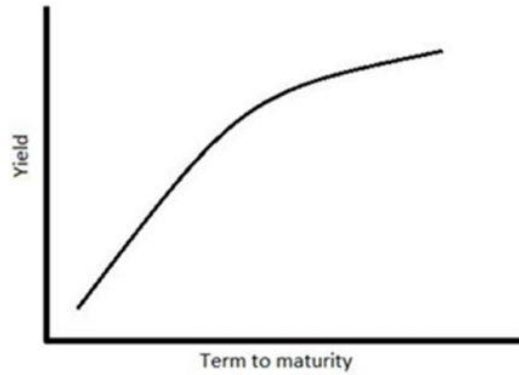
Will the borrower default?

Bet on Credit Risk

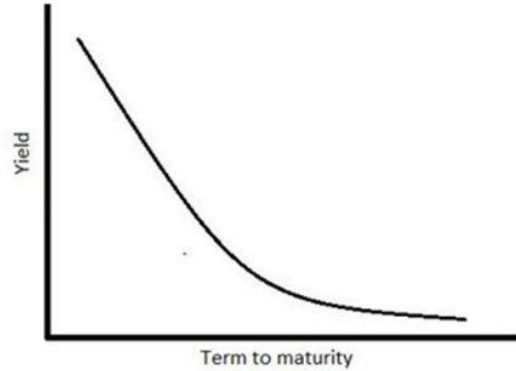
Borrower-specific

Yield Curve

Normal Yield Curve



Inverted Yield Curve



Short-end, Belly, Long-end Rates

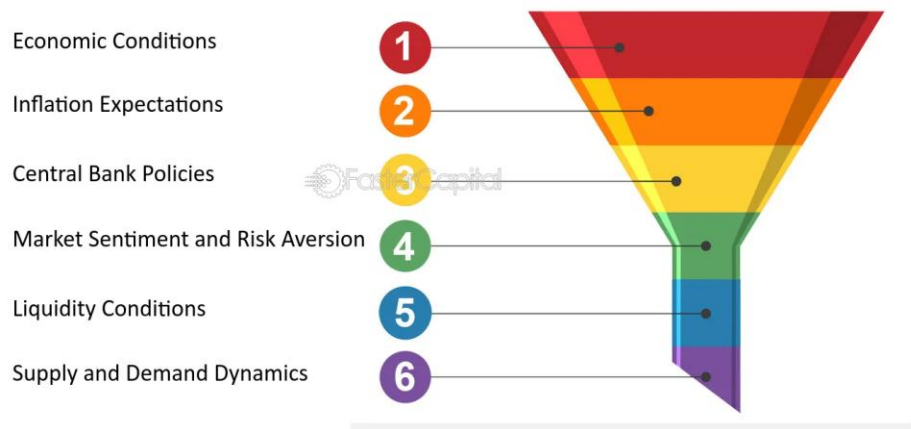
- Interest rates essentially make you decide whether you save or spend money, which effectively affects economic activity
- Higher interest rates means more expensive borrowing and more attractive saving, so economy cools as folks spend less, business invest less into projects
- US issues bonds up to 30 year maturity, each bucket of maturity is affected by different factors
 - Short-end (1-month to 2-year): current and immediate path of policy rate (what's the Fed doing today?)
 - Belly (3- to 7-year): intermediate path of policy rate and business cycle (when will the Fed pivot?)
 - Long-end (10-year to 30-year): long-term economic prospects (what's the long-term inflation/growth?)

Longer you go, the bond is more influenced by TP

Long-term rate = expected short-term rates + term premium

I can roll a series of short-term rates into a long-term rate, but it's not guaranteed the rates don't change! TP is my insurance!

Factors Influencing Term Premium



Post-08 changes in monetary policy

History of US interest rates



Cutting rates below 0?

- Great Recession after GFC
- Fed cut rates from 5.50% to 0.50%
- You can't cut rates below 0%
 - Why?

Cutting rates below 0?

- Great Recession after GFC
- Fed cut rates from 5.50% to 0.50%
- You can't cut rates below 0%
 - Why?
 - It means if you deposit money at banks, you have to pay for it!
 - People will have to withdraw money from their bank accounts
 - Monetary policy works by influencing savings and investment decisions, **no savings – monetary policy will no longer work!**

So what did the Fed do instead?

- Quantitative Easing (QE): buying long-term government bonds
 - I (Fed) will pay you so you can spend money on things rather than save!
 - Conducting monetary policy on the long-end, not the short-end
- A Regime of ample reserves

Then, inflation hit in 2022

- Fed did the reverse – hiking rates + QT
- Ample reserves → scarce reserves
- Fed ended QT in December 2025

In a regime of scarce reserves, short-term interest rates are more volatile (banks are more sensitive to changes in STIR)...

Different “bets” in rates markets

What are the different possible ways to trade rates?

- **Directional bet:** higher vs. lower rates?
- **Curvature bet:** steeper vs. flatter yield curve?
- **Basis bet:** will there be a difference between cash and its derivative?
- **Breakeven bet:** will inflation be higher than markets expect?

Directional bets

- Buy bonds across the board
- Raise the duration of my portfolio

Curvature bets

Bull

Bear

Steeper

Flattener

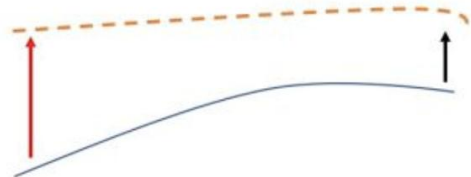
Curvature bets

- Spots & Curve trades – steepeners, flatteners. E.g. 2s10s steepener trade

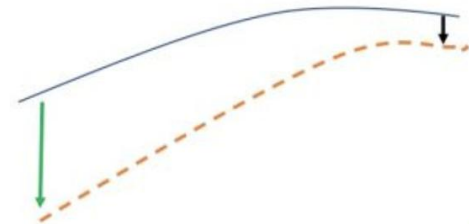
Bull Flattening



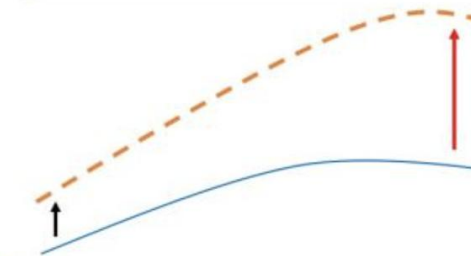
Bear Flattening



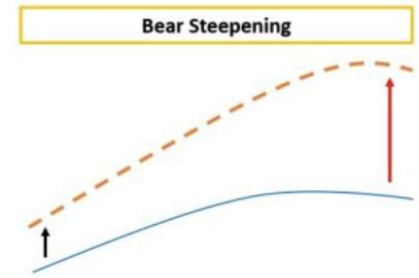
Bull Steepening



Bear Steepening



Curvature bets



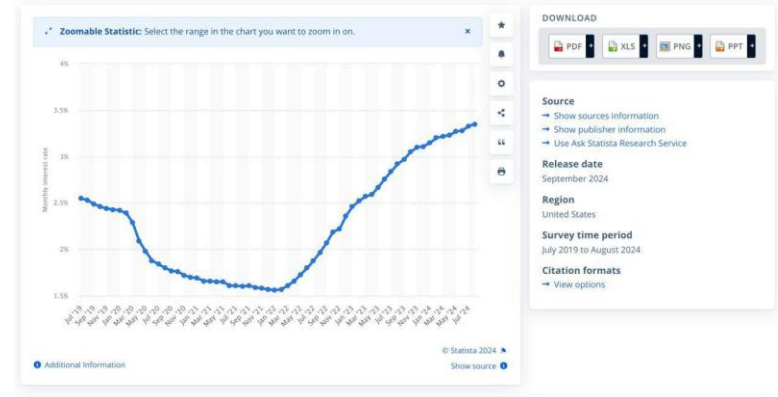
Bear Steepener Trade in the US

- Interest rates will increase by a much larger extent in the long end compared to the short-end
- ie Short long term treasury bond (20 years) and long short term treasury bond (2 years)

Thesis:

- Short term rate cuts are already priced in by market
- Long term inflationary pressures, rising deficits (increasing issuance) in the US
- Rising interest rates on US debt will make it more difficult
 - For cheap borrowing and long term rates/yield curve will
 - Remain elevated

Average monthly interest rates on the United States total interest-bearing debt from July 2019 to August 2024



Basis trade

Basis: difference in payoff between the cash security and its derivative security

It arises because of market frictions (i.e. it's not normal)

Since the late 2010s, US has been expanding fiscal deficit → more supply for treasuries

At the same time, banks (traditional market makers in bonds) are heavily regulated → less demand for treasuries

NBFIs have taken on increasing role in treasury market making, but they come with **very different incentives**

More supply + less demand → less liquidity + more market frictions

Basis trade – Real Money

1. Client gives me
\$10m to manage
for 10 years

2. In 10 years'
time, I need to
generate X% return

3. At the end, I get
Y% of my AUM

Basis trade – Fast Money

1. Client gives me
\$10m to manage
for 1 quarter/month

2. In 1
quarter/month, I
need X% return

3. At the end, I get
Y% of my AUM +
Z% performance
fee

Basis trade



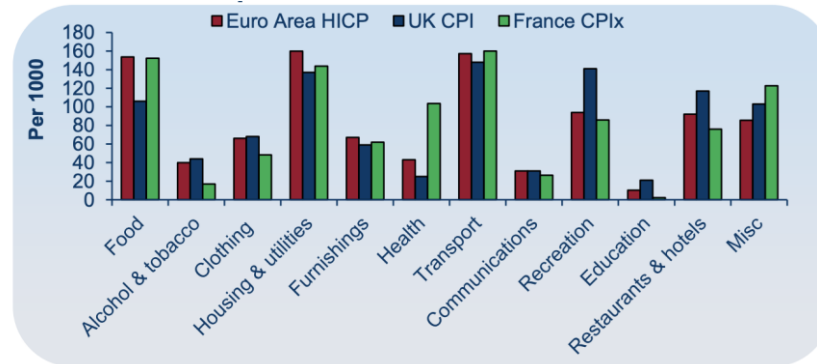
Basis trade – volatile!

Basis is typically very small (a few bps)

Hedge funds employ leverage via repo with banks, collateralising the bonds they buy

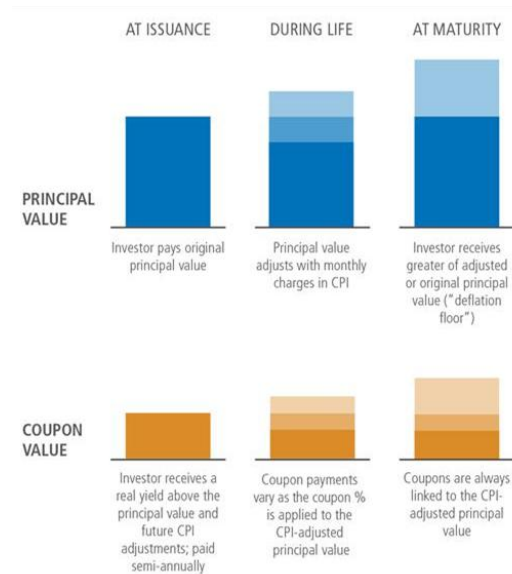
Breakeven bet

- Inflation linked products are a type of indexed product, meaning cash flows depend on an index
 - In the UK, the RPI is commonly used
 - Note: RPI is changing *soon*. Index will soon become like the CPIH from 2030
 - In the USA, CPI is more commonly used
- Inflation linked products are used to commonly hedge large corporate inflation exposure
 - i.e. Pension pay-outs that increase with an inflation index



Breakeven bet

- Inflation Linked Government Bonds are similar to vanilla (normal) government bonds where you have incremental, *predetermined* cashflows. But the difference here is that both the coupons and the face value are scaled in accordance with the inflation index (specified for the product)
 - Why would I ever buy a nominal bond if I can just always be protected from inflation with these?
- Inflation Swaps
 - Similar to normal IRS but instead the referenced rate is an inflation rate
 - Usually *zero-coupon*
 - Keep an eye out for the 5y5y



Source: PIMCO. Sample for illustrative purposes only

Breakeven Inflation Rate

- The difference between the yield on the nominal and linker is known as the *breakeven inflation rate*
 - This can be seen as market expectations of future inflation rates